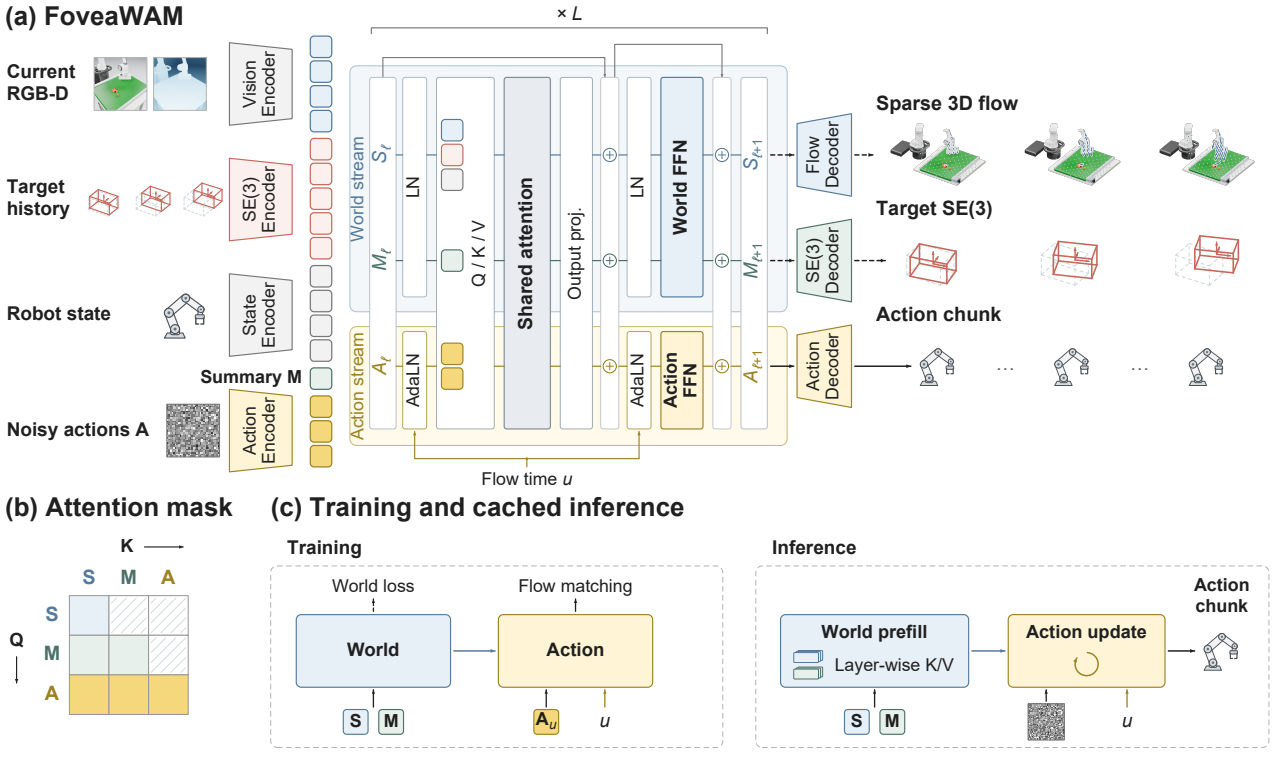

(a) FoveaWAM
× L
Current RGB-D
Vision Encoder
Target history
SE(3) Encoder
Robot state
State Encoder
Summary M
Noisy actions A
Action Encoder
World stream
Action stream
LN
AdaLN
Q / K / V
Shared attention
Output proj.
LN
World FFN
AdaLN
Action FFN
Flow time u
Flow Decoder
SE(3) Decoder
Action Decoder
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
K
Q
S
M
A
(c) Training and cached inference
Training
World loss
Flow matching
World
Action
Inference
World prefill
Layer-wise K/V
Action update
Action chunk